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Research Statement

Deferral-Aware Clinical Foundation Models for Longitudinal Diagnostic Reasoning

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Research vision

My research asks how clinical AI systems should collaborate with human experts when decisions are uncertain, information is incomplete, and responsibility cannot be handed over casually. Rather than treating healthcare AI as either autonomous prediction or passive decision support, I design systems that allocate responsibility. These systems should act on routine cases, ask for missing information, communicate uncertainty, escalate urgent situations, and defer difficult cases to the right clinician. During my DPhil, I developed learning-to-defer and human-AI collaboration methods for healthcare, including guided deferral with large language models, deferral to unseen experts, and coherent handoffs for structured medical decisions [1–3]. As a postdoctoral scholar in the Fries Lab, I would extend this work to longitudinal multimodal foundation models for healthcare, with a focus on dynamic deferral, expert-facing evaluation, and data-centric feedback loops.

The central gap is that clinical foundation models are still often evaluated as answer generators. Given a vignette, note, image, or patient trajectory, the model is asked to produce a diagnosis, summary, risk score, or recommendation. Real clinical reasoning is more interactive. In a diagnostic consultation, the right next step may be to ask a focused question, request an image or laboratory test, retrieve prior history, compare the current trajectory to similar patients, seek a specialist opinion, safety-net, or escalate urgently. Recent work on AMIE shows the promise of LLM-based diagnostic dialogue and differential diagnosis assistance, while also showing that clinical utility cannot be reduced to single-turn accuracy [4, 5]. My proposal is to make deferral a dynamic part of consultation rather than a final reject option. A clinically useful model should reason not only about what is likely, but also about which action would most improve the safety and utility of the human-AI team.

This framing aligns with the Fries Lab’s work on foundation models for longitudinal healthcare data, EHR benchmarking, clinician-generated instruction-following datasets, time-to-event modelling, long-context clinical prediction, and cross-site adaptability [6–10]. The lab’s emphasis on realistic clinical data is essential because dynamic deferral depends on noisy and sparse records, shifting populations, heterogeneous clinicians, and tacit workflow knowledge.

Preparation

My prior research provides the foundation for this research. In my AAAI 2025 paper, I developed a guided deferral system in which an instruction-tuned LLM parses clinical reports, handles confident cases autonomously, and defers uncertain cases to humans with textual guidance. A pilot study showed that guided humans outperformed both unguided humans and the LLM alone, which supports the view that deferral should improve team performance rather than simply transfer errors downstream [1]. This work also addressed constraints that matter in healthcare AI, including privacy, open-source deployment, modest compute, and calibration under imbalanced clinical data.

A second strand of my work studies how deferral policies adapt when the available expert changes. In *Identity-Free Deferral*, I introduced a role-indexed architecture for deferring to unseen experts, using few-shot evidence to estimate expert competence without relying on brittle class-identity shortcuts [2]. This is directly relevant to clinical environments where staffing, specialty expertise, and site-specific practice vary. A third strand studies structured handoffs. In coherent hierarchical multi-label deferral, I formalised cases where independent label-wise deferral can produce incoherent clinical handoffs, then developed contract-aware methods that preserve taxonomic and workflow coherence [3]. Finally, my scoping review of human-AI collaboration in healthcare identified a field-level gap. Studies often measure short-term task performance, while fewer evaluate workflow impact, calibrated reliance, accountability, or longitudinal clinical outcomes [11]. Together, these projects prepare me to contribute both technical methods and evaluation frameworks for expert-facing clinical foundation models.

Proposed postdoctoral research

Aim 1 Dynamic deferral policies for diagnostic dialogue

I will develop deferral-aware diagnostic dialogue systems in which the model's action space extends beyond “predict” versus “defer”. At each turn, a clinical foundation model should choose among actions such as asking a focused history question, requesting missing EHR context, recommending an image or laboratory test, retrieving comparable patient trajectories, presenting a provisional differential diagnosis, providing safety-netting information, or escalating to a clinician. The technical problem is a sequential decision problem under uncertainty, where the objective is team utility rather than standalone model accuracy. Learning-to-defer provides the formal starting point [12,13]. CoDoC and AMIE provide complementary clinical precedents, with CoDoC showing how AI can route cases to a clinical workflow and AMIE showing the potential of multi-turn diagnostic reasoning with structured evaluation [4,5,14].

Methodologically, I would formulate diagnostic dialogue as a partially observed, multimodal decision process over patient state, available evidence, model uncertainty, and expert capacity. The model would maintain calibrated beliefs over diagnoses, urgency, and missing information. A deferral policy would select the next action according to expected information gain, risk, cost, and the estimated competence of available human experts. This generalises deferral from a final handoff into a repeated consultation action. It also connects to longitudinal foundation models because the decision to ask, retrieve, or defer should depend on the patient's temporal trajectory rather than only the current text turn.

A first implementation could use simulated patient encounters grounded in real or semi-synthetic EHR trajectories, followed by retrospective evaluation and clinician review. Extensions would incorporate visual or tabular evidence, making the system decide when to request or interpret imaging, labs, notes, medications, or prior outcomes. This work would produce a methodological paper on dynamic deferral for multimodal diagnostic dialogue, a reusable ask-retrieve-defer evaluation environment, and a clinician or simulation study comparing static answer generation, terminal deferral, and dynamic deferral.

Aim 2 Benchmarks and rubrics for multi-step human–AI clinical reasoning

Dynamic deferral requires new evaluation. Top- k diagnostic accuracy is useful but incomplete. An unsafe model can reach the right diagnosis after asking unnecessary questions, missing urgent escalation, anchoring clinicians on a misleading differential, or failing to surface uncertainty. I will build benchmarks and rubrics that evaluate expert-facing foundation models as collaborators in multi-step reasoning, extending the spirit of AMIE's clinician- and patient-centred rubrics while adapting them to longitudinal EHR-grounded workflows [4,5].

The benchmark would score not only final diagnosis, but also intermediate actions. It would assess whether the model asked clinically relevant questions, identified missing critical information, requested appropriate modalities, escalated urgent presentations, communicated uncertainty, and handed cases to clinicians with useful guidance. I would emphasise temporal and distributional stress tests, including sparse histories, low-prevalence conditions, site shifts, changing disease prevalence, incomplete notes, and evolving patient trajectories. This fits naturally with Fries Lab resources and publications such as MedAlign, EHRSHOT, MOTOR, and long-context EHR evaluation [6–8,10].

I would develop a tiered evaluation pipeline. The first tier would use retrospective EHR and case-based tasks with expert labels for missing information, urgency, and deferral appropriateness. The second tier would use simulated consultations to test policies over multiple turns. The third tier would use human-subject or clinician-reader evaluations to measure reliance, workload, override behaviour, and handoff quality. A core contribution would be a set of metrics for *calibrated collaboration*, meaning that clinicians rely on the model when it is likely to help, contest it when it is likely to be wrong, and remain cognitively engaged rather than passively verifying outputs [11].

Aim 3 Data-centric feedback loops from deferral, correction, and expert disagreement

Deferral is not only a safety mechanism. It is also a data-generating mechanism. Every handoff can reveal what the model did not know, which evidence was missing, how clinicians corrected or reframed the case, and which tacit workflow considerations mattered. I will study how these signals can be converted into supervision for improving clinical foundation models under noise, sparsity, and domain shift.

This aim would treat clinician interaction traces as structured feedback, including questions asked, evidence requested, disagreements, overrides, final decisions, and follow-up outcomes. I would develop methods to value and curate these signals, distinguishing high-value expert feedback from noisy or context-dependent variation. This connects to data-centric AI and clinician-generated datasets such as MedAlign, while my work on unseen-expert deferral would help model heterogeneity across specialists and sites [2,6]. A concrete project would train a deferral-aware clinical foundation model using interaction-derived supervision. Cases frequently deferred for missing history could improve question-asking policies. Specialist corrections could improve routing and attribution. Delayed-diagnosis or escalation cases could become safety stress tests. The goal is to close the loop between deployment-aware evaluation and model improvement, while preserving clinician accountability and monitoring for drift, bias, and over-reliance.

Fit and trajectory

The Fries Lab is an ideal setting for this programme because it combines foundation-model research with hospital data science, longitudinal multimodal EHRs, clinician-generated datasets, benchmarking, and human–AI teaming. I bring expertise in learning-to-defer, guided LLM decision support, expert heterogeneity, structured clinical handoffs, and human–AI evaluation. I also bring practical experience from NHS data science, including modelling on real healthcare datasets and working with clinical and operational stakeholders.

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